

EcoWaste AI

Using Artificial Intelligence to Reduce Environmental Waste

Done by: AMIRTHA GANESH R

October 18, 2025

Project Overview

EcoWaste AI is an AI-driven sustainability project that combines computer vision, machine learning, and data analytics to help users make smarter, greener choices.

The application allows users to upload a photo of a waste item, which is automatically classified as either *Organic* or *Recyclable*, and then predicts how much CO₂ could be saved if properly recycled or composted. Additionally, the system provides personalized sustainability tips and an optional user profile that clusters recycling habits to deliver insights on improving environmental impact.

Core Functionality: User uploads a photo of waste → AI classifies it → Calculates CO₂ savings → Provides recycling guidance.

Problem Statement

Waste management represents a critical environmental challenge. Annually, millions of tons of recyclable and compostable materials end up in landfills, contributing to:

- Increased CO₂ and methane emissions
- Ground and water pollution from toxic leachate
- Wildlife consumption of toxic waste
- Air pollution from incineration

While most individuals desire to sort waste responsibly, many lack the knowledge or daily motivation to do so. This project leverages AI to bridge that gap.

Project Objectives

1. Classify waste into Organic and Recyclable categories using computer vision
2. Estimate CO₂ savings (in kg) for properly processed items
3. Cluster user behavior to identify recycling patterns and generate personalized recommendations
4. Deploy an interactive web application using Streamlit that runs on standard laptops without requiring GPU resources
5. Promote sustainability awareness through data-driven insights and machine learning

Methodology

Step 1: Data Collection

Dataset Source: Waste Classification Data from Kaggle

The dataset comprises approximately 25,000 annotated images divided into two classes:

- **Organic (O):** Food scraps, leaves, and biodegradable items
- **Recyclable (R):** Plastics, glass, metals, paper, and cardboard

Dataset Split:

- Training: 22,564 images (80%)
- Testing: 2,513 images (20%)

Step 2: Data Preprocessing

- Loaded images using `tf.keras.utils.image_dataset_from_directory()`
- Resized all images to 160x160 pixels (optimized for MobileNetV2)
- Normalized pixel values to range `[-1, 1]` using `tf.keras.applications.mobilenet_v2.preprocess_input()`
- Split dataset: 80% training, 20% validation

Step 3: Image Classification Model

Architecture: Transfer Learning with MobileNetV2 ($\alpha = 0.35$) pre-trained on ImageNet

```
base_model = MobileNetV2(
    weights='imagenet',
    include_top=False,
    input_shape=(160,160,3),
    alpha=0.35
)
base_model.trainable = False

x = base_model.output
x = tf.keras.layers.GlobalAveragePooling2D()(x)
x = tf.keras.layers.Dropout(0.3)(x)
output = tf.keras.layers.Dense(2, activation='softmax')(x)
model = tf.keras.Model(inputs=base_model.input,
                       outputs=output)
```

Training Parameters:

- Optimizer: Adam (learning rate = 0.0001)
- Loss Function: Categorical Crossentropy
- Batch Size: 32
- Epochs: 4

Results:

Metric	Train	Validation	Test
Accuracy	91.4%	92.9%	88.8%
Loss	0.22	0.18	0.16

Key Finding: MobileNetV2 achieved excellent performance with minimal dataset requirements and computational overhead, making it ideal for edge deployment.

Step 4: CO₂ Savings Regression Model

Following classification, each item’s CO₂ savings is predicted based on material type and weight.

Model: RandomForestRegressor

Features:

- Material type (Organic/Recyclable)
- Weight (user input in kg)

CO₂ Factors:

Material Type	CO ₂ Saved per kg
Organic	0.5 kg CO ₂
Recyclable	1.5 kg CO ₂

Model Performance:

- Mean Squared Error (MSE): 0.02
- R² Score: 0.96

Step 5: User Behavior Clustering

Objective: Understand user recycling patterns and generate actionable insights

Model: K-Means Clustering (k=3)

Features:

- Total weight of items uploaded
- Average CO₂ saved per item
- Fraction of organic uploads
- Fraction of recyclable uploads

Identified User Clusters:

Cluster	Behavior	Recommendation
Recycle-Heavy	Mostly recyclable items	Switch to reusable products
Organic Recycler	Mostly compostable items	Continue composting regularly
Mixed Recycler	Balanced pattern	Track weekly CO ₂ savings

Step 6: Web Application Deployment

Framework: Streamlit

Application Workflow:

- 1. User uploads an image of waste
- 2. User enters item weight (in kg)
- 3. AI predicts: waste category, confidence level, and estimated CO₂ saved
- 4. App displays: classification results, personalized tips, and environmental impact metrics

Technologies Used:

- Streamlit (Frontend UI and interactive dashboard)
- TensorFlow and Keras (Computer Vision model)
- Scikit-learn (Regression and clustering models)
- Joblib and JSON (Model serialization)
- PIL (Image preprocessing)

Results Summary

Model Component	Metric	Score
MobileNetV2 Classifier	Test Accuracy	88.8%
RandomForest Regressor	R ² Score	0.96
K-Means Clustering	Silhouette Score	0.45
Overall Application	Avg. Inference Time	<2 seconds

Example Predictions

Example 1: Banana Peel

- Predicted Class: Organic
- Confidence: 96%
- Estimated CO₂ Saved: 0.45 kg
- Insight: Composting prevents methane emissions

Example 2: Plastic Bottle

- Predicted Class: Recyclable
- Confidence: 92%
- Estimated CO₂ Saved: 1.25 kg
- Insight: Clean and dry before recycling for optimal reprocessing

Technology Stack

Category	Technologies
Programming Language	Python
Machine Learning	TensorFlow, Keras, Scikit-learn
Data Processing	Pandas, NumPy
Frontend	Streamlit
Visualization	Matplotlib, Seaborn
Development Environment	Google Colab, VS Code
Model Explainability	SHAP (optional)

Key Learnings

- Integrated Computer Vision and Tabular Machine Learning within a unified framework
- Leveraged transfer learning to achieve high accuracy with limited computational resources
- Constructed a complete ML pipeline: data preprocessing, classification, regression, clustering, and production deployment
- Designed responsive and intuitive user interfaces using Streamlit with CSS customization
- Demonstrated practical applications of AI in climate action and sustainable living

Conclusion

EcoWaste AI demonstrates how artificial intelligence can empower individuals to make environmentally conscious decisions. By combining state-of-the-art deep learning with user-centric design, this project bridges the gap between sustainability awareness and actionable behavior change. The lightweight architecture ensures accessibility for users worldwide, making environmental impact quantifiable and personal.

Impact: *Together, every recycled item and composted material contributes to a more sustainable future.*